

Submission A solution to Problem 7

A counterexample

The answer is no. We give a counterexample in $G = \langle a, b, c \rangle$. Write $\bar{x} = x^{-1}$, and set

$$w = \bar{a} a b \bar{b} \bar{a} a c \bar{c} \bar{a} a.$$

This word is freely trivial, by deleting the adjacent inverse pairs

$$\bar{a}a, \quad b\bar{b}, \quad \bar{a}a, \quad c\bar{c}, \quad \bar{a}a.$$

It is quasi-reduced: the only adjacent inverse pairs are the five displayed ones, and in each case the following letter, if there is one, is not the first letter of that pair.

We shall use two elementary consequences of the face condition in the definition. Compactify the upper half-plane to a disk and put the marked boundary points in cyclic order. For a matching M , let $\Gamma(M)$ be the graph obtained from M by making each crossing a vertex; the original boundary endpoints are leaves.

Lemma 1. *For every admissible matching M , the graph $\Gamma(M)$ is a forest.*

Proof. Suppose that $\Gamma(M)$ contains a cycle. Boundary vertices have valence one, so every cycle is contained in the interior of the disk. Choose one which is innermost: the disk bounded by it contains no other cycle. It in fact contains no edge of $\Gamma(M)$. Indeed, any component of the graph lying in that disk either contains a cycle, which would be smaller, or is a tree. Such a tree cannot be wholly interior, because it would have an interior leaf; and if it attaches to the chosen cycle in two places, the path between the attachment points gives a smaller cycle. Thus it would have to attach in exactly one place, again forcing an interior leaf. This is impossible since all interior vertices have valence four and all leaves of $\Gamma(M)$ lie on the boundary of the disk.

Thus one complementary region has this cycle as a boundary component. The corresponding component of $M \cap \partial\bar{R}$ is closed and has no boundary endpoint. This contradicts the defining requirement that every such component have boundary consisting of two marked boundary points with inverse labels. $\square$

Lemma 2. *Suppose that a letter x occurs in the word exactly once and x^{-1} occurs exactly once. Then, in every admissible matching, the two corresponding endpoints are joined by an uncrossed interval.*

Proof. Let the two endpoints be p and q . At p , the two local sides of the incident edge belong to two components of face boundary. By the label condition, both of these components must end at q , since q is the only endpoint with the inverse label.

By the preceding lemma the component of $\Gamma(M)$ containing p is a tree, and hence there is a unique graph path from p to q . If this path met a crossing, let c be the first crossing encountered when travelling from p . At c , the two local face-boundary sides of the incoming edge leave along two different half-edges. Removing c from the tree separates those two half-edges into different components, so at most one of the two resulting face-boundary arcs can continue to q . This contradicts the previous paragraph. Therefore the path from p to q has no crossing, and is a single uncrossed interval. $\square$

In the word w , the letters $b, \bar{b}, c, \bar{c}$ each occur only in the displayed adjacent inverse pairs. Hence every admissible matching contains the uncrossed intervals $(3, 4)$ and $(7, 8)$. Since their endpoints are adjacent on the boundary, these intervals cut off empty boundary disks. Deleting them and renumbering the remaining endpoints 1, 2, 5, 6, 9, 10 gives a cell-preserving isomorphism from F_w to the analogous six-endpoint complex for

$$u = \bar{a} a \bar{a} a \bar{a} a.$$

Here u is only an auxiliary labelled boundary word; the same matching definition is being used after the forced intervals have been removed.

We now compute this six-endpoint complex. For u , odd endpoints have label $\bar{a}$ and even endpoints have label a . Write

$$ij|kl|mn$$

for the pairing $\{i, j\}, \{k, l\}, \{m, n\}$. By Lemma 1, two intervals cannot cross twice, since two crossings between them would make a bigon cycle. Also, all three pairs of intervals cannot cross, since the three crossings would form a triangular cycle. Hence the crossing pattern is determined by the endpoint pairing: two pairs cross exactly when their endpoints interleave around the boundary. There is no further isotopy datum here; a regular neighbourhood of the resulting planar forest is unique once the cyclic order of its six boundary leaves is fixed.

Conversely, take any pairing for which the interleaving graph is a forest, and draw it with one transverse crossing for each interleaving pair and no other crossings. Then $\Gamma(M)$ is a forest. Every component of $M \cap \partial \bar{R}$ is therefore an arc joining two boundary endpoints. Push such an arc slightly into the adjacent face. The pushed-off arc is disjoint from the diagram and cuts the disk into two pieces. No interval can have its endpoints on different sides, so each side contains an even number of marked endpoints. Therefore the two endpoints of the face-boundary arc have opposite parity. For the alternating word u , opposite parity is exactly the inverse-label condition. Thus precisely the pairings with forest interleaving graph are admissible. Among the fifteen pairings of six points, the only excluded one is the all-crossing pairing 14|25|36.

The resulting cells are the following. The five vertices are

$$\begin{aligned} v_0 &= 12|34|56, & v_1 &= 12|36|45, & v_2 &= 14|23|56, \\ v_3 &= 16|23|45, & v_4 &= 16|25|34. \end{aligned}$$

The six edges are

$$\begin{aligned} e_{01} &= 12|35|46, & e_{02} &= 13|24|56, & e_{13} &= 13|26|45, \\ e_{23} &= 15|23|46, & e_{04} &= 15|26|34, & e_{34} &= 16|24|35, \end{aligned}$$

where the subscripts indicate the two endpoint vertices. The three squares are

$$Q_{12} = 13|25|46, \quad Q_{14} = 14|26|35, \quad Q_{24} = 15|24|36,$$

and their boundaries, obtained by the two local resolutions of each crossing, are

$$\begin{aligned} \partial Q_{12} &= e_{01} \cup e_{13} \cup e_{23} \cup e_{02}, \\ \partial Q_{14} &= e_{01} \cup e_{13} \cup e_{34} \cup e_{04}, \\ \partial Q_{24} &= e_{02} \cup e_{23} \cup e_{34} \cup e_{04}. \end{aligned}$$

Thus the one-skeleton is the theta graph with poles v_0 and v_3 and with three length-two paths through v_1, v_2, v_4 . The square Q_{12} fills the cycle formed by the paths through v_1 and v_2 . The square Q_{14} fills the cycle formed by the paths through v_1 and v_4 . These two squares are glued along the common path through v_1 , so their union is a disk whose boundary is the remaining cycle, the one through v_2 and v_4 . Finally Q_{24} is glued along exactly this boundary. Hence the whole complex is a 2-sphere:

$$F_w \cong S^2.$$

In particular F_w is not contractible.